

Synthesis of Isobutane from n-Butane via Conventional Distillation – Aspen HYSYS Simulation

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Background & Description:

This study focuses on simulating and analyzing the process of separating isobutane (i-butane) from n-butane, using Aspen HYSYS as the process simulation tool. Isobutane is a critical component in alkylation units for producing high-octane gasoline. Since crude oil contains insufficient isobutane, it is essential to recover it efficiently from mixed hydrocarbon streams containing both isomers.

The process modeled is inspired by the configuration discussed in Luyben (2011), but unlike the original setup which includes a side-reactor loop, the present simulation explores a pure distillation-based approach. The goal was to assess whether a high reflux ratio alone could achieve the required separation of i-butane and n-butane, without requiring a reactive section.

Process Description:

The simulated flowsheet in Aspen HYSYS involves a **single conventional distillation column** with the following specifications:

- **Feed Stream:** A mixture of
 - 1 mol% propane (C3)
 - 20 mol% isobutane (i-C4)
 - 74 mol% n-butane (n-C4)
 - 5 mol% isopentane (i-C5)is fed to a **62-stage column** at **stage 30**, under **90 psia (~6.2 bar)** and at a temperature around **61.2 °C**.
- **Column Operation:**
 - **Top Product (Distillate):** Rich in isobutane (target purity $\geq 90\%$)
 - **Bottom Product:** Containing n-butane and isopentane (target i-C5 purity $\geq 95\%$)
 - **Reflux Ratio:** Set to **21** to enhance the separation between i-butane and n-butane
 - **Condenser:** total $\sim 48^\circ\text{C}$
 - **Reboiler Temperature:** $\sim 96^\circ\text{C}$
- **Conversion Reactor (adiabatic):**
 - **Input: Side-draw vapor from stage 45**
 - Temperature: 334.36 K (61.2 °C)
 - Pressure: 642,270 Pa (6.42 bar)
 - Flowrate: 27.7 mol/s (~ 99.72 kgmol/h)
 - The vapor is **pumped and heated** to **95.1°C (368.2/K)** under **861.8 kPa**, then sent to a **Conversion Reactor**
 - Reaction:
 - $\text{n-C4} \rightarrow \text{i-C4}$
 - Conversion rate: 38%
 - Duty: Set to 0 to simulate adiabatic conditions
 - This configuration mimics partial isomerization before fractionation, improving the isobutane yield in the distillate

Thermodynamic package: The Peng-Robinson equation of state was used throughout the simulation, due to its suitability for light hydrocarbon systems and moderate pressures. This model allows accurate representation of phase equilibria between closely boiling compounds like i-butane and n-butane.

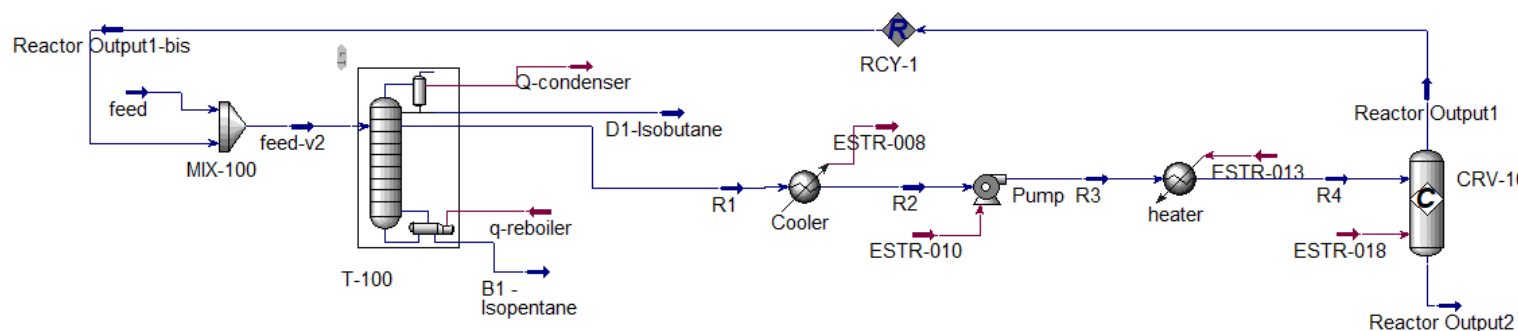
Conclusion: This Aspen HYSYS simulation demonstrates that combining a **vapor side draw**, an **adiabatic conversion reactor**, and a **high-reflux-ratio distillation column** provides an efficient and relatively simple configuration for producing high-purity isobutane from n-butane.

The model replicates key aspects of the design presented by Luyben (2011), achieving both **significant isomerization (38%)** and effective separation — with the **reflux ratio of 21** proving sufficient to meet product purity specifications, even for components with close boiling points like i-butane and n-butane.

This approach eliminates the need for a complex recycle loop while ensuring acceptable removal of heavier components (such as i-pentane) in the bottom stream. It offers a **simplified and economically attractive alternative** to traditional column-reactor systems, particularly in cases where process intensification, equipment minimization, or operational simplicity are critical.

The results validate the use of **Aspen HYSYS** as a reliable tool for simulating **thermodynamically challenging separations**, and emphasize the importance of key parameters such as **reflux ratio**, **side draw location**, and **conversion extent** in designing integrated reactive-separation processes.

Flowsheet



Results:

Master Property Tableau

Object	Feed	Reactor Output 1-bis	Feed-V2	B1-Isopentane	D1-Isobutane	R1	R4	Reactor Output2
T (°C)	25.00	95.92	67.07	93.32	46.59	60.31	73.89	95.93
P (kPa)	101.3	861.8	101.3	650.9	620.5	642.4	861.8	861.8
Molar Flow (kgmole/h)	98.87	202.7	301.6	5.234	93.80	202.6	202.6	0
Mole Fract (i-butane)	0.2000	0.3418	0.2953	0	0.9094	0.0185	0.0185	<0
Mole Fract (i-Pentane)	0.0500	0.0029	0.0183	0.9500	0	0.0027	0.0027	<0
Mole Fract (n-Butane)	0.7400	0.6554	0.6831	0.0500	0.08	0.9788	0.9788	<0
Mole Frac (Propane)	0.01	0	0.0033	0	0.0105	0	0	<0

Nb : Insignificant quantities (<0) of components in the product streams can be considered negligible, effectively approximating them to zero without impacting the overall process results or conclusions (see flowsheet).

References: Luyben, W. L. (2011). *Control of an Isomerization Column/Reactor Process*. *Industrial & Engineering Chemistry Research*, 50(6), 3382–3389. <https://doi.org/10.1021/ie101724k>